

datadriftR: An R Package for Concept Drift Detection in Predictive Models

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Abstract Predictive models often degrade due to data drift, particularly concept drift, where the relationship between variables and the response changes. Traditional detection methods based on accuracy or variable distributions often miss subtle concept drifts. This paper presents, an R package for detecting concept drift and introduces a novel method, Profile Drift Detection (PDD). PDD leverages explainable artificial intelligence tool Partial Dependence Profiles (PDPs) to detect and explain drift causes. It quantifies PDP changes using innovative metrics, balancing sensitivity and computational efficiency. Aligned with MLOps practices, this approach focuses on model monitoring and adaptive retraining in dynamic environments. Experiments on synthetic and real-world datasets show that PDD outperforms existing methods, maintaining high accuracy while ensuring stability. The results demonstrate its robustness for real-time applications and ability to adapt models effectively. The paper discusses its advantages, limitations and potential extensions for broader applications.

1 Introduction

Predictive models assume train and test data are independent and identically distributed (Cieslak and Chawla, 2009) but this rarely holds, reducing performance (Alaiz-Rodríguez and Japkowicz, 2008). As data evolves, models must adapt (Moreno-Torres et al., 2012) while ensuring accuracy and efficiency (Korycki and Krawczyk, 2023). Model monitoring, a key MLOps component, maintains ML consistency in real-world applications (Figure~1) (Biecek, 2019; Mougán and Nielsen, 2023).

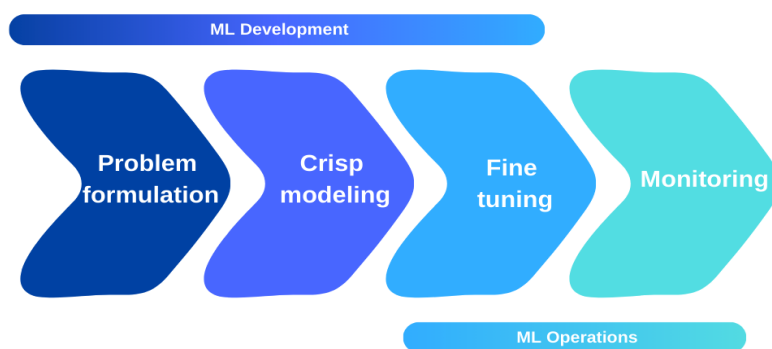


Figure 1: The predictive modeling process

Data drift, classified into covariate, label and concept drift, disrupts model behavior. It appears in scenarios like pandemic-driven electricity demand (Fumagalli et al., 2023), airline customer shifts (Garg et al., 2021) and supply chain changes (Kulinski and Inouye, 2023). Timely detection is crucial for model reliability.

Data drift detection methods fall into statistical, sequential and window-based categories. Concept drift, a change in explanatory-response variable relationships, often eludes traditional methods (Demšar and Bosnić, 2018; Duckworth et al., 2021; Rahmani et al., 2023). This underscores the need for MLOps, which integrates model monitoring and adaptation to maintain performance in dynamic environments. A key challenge arises when concept drift occurs without shifts in performance or distribution metrics, making conventional detection unreliable and necessitating alternative approaches.

Demšar and Bosnić (2018) highlighted that existing drift detection methods rely on performance and distribution metrics, which fail to explain underlying model changes. They proposed a method using local interpretable model-agnostic explanations (Strumbelj et al., 2009), showing improved detection. Duckworth et al. (2021) used normalized SHAP (Lundberg and Lee, 2017) to detect drift in emergency department models post-COVID-19.

Muschalik et al. (2022) developed a concept drift detection approach using permutation-based variable importance (Breiman, 2001), emphasizing the need for improvements in complexity, sensitivity and explainability. Rahmani et al. (2023) found SHAP effective for variable importance but insufficient for distinguishing covariate and concept drift. Mougán and Nielsen (2023) warned that statistical

tests rely on data distribution alone, causing false alarms and proposed an explainable method using bootstrap uncertainty estimation and SHAP.

Mattos et al. (2021) addressed the lack of explainability in concept drift detection by developing a decision tree-based technique but noted its limitations, including high training costs and difficulty tracking variables in non-tree models. Fumagalli et al. (2023) extended permutation-based variable importance for incremental models but stressed the need to improve efficiency, complexity and sensitivity.

XAI-based data drift detection methods are limited and often rely on high-cost approaches like permutation-based importance or SHAP, which are not always practical for detecting changes in response-explanatory variable relationships. To address this, we propose a solution using Partial Dependence Profiles (PDP), a versatile XAI tool for examining response-explanatory relationships, hyper-parameter optimization (Moosbauer et al., 2021) and variable importance analysis (Ingilis et al., 2022).

Software implementing complex drift detection methods is also scarce. R tools like **vetiver** and **pins** monitor model metrics but lack dedicated drift detection. **drifter** detects drift via distribution distances and PDPs but is limited to comparing trained models, while **harbinger** focuses on time series anomalies. In contrast, Python offers advanced tools such as EvidentlyAI¹, SeldonIO², NannyML³, Frouros⁴ and River⁵.

To bridge these gaps, we introduce **datadriftR**, an XAI-based concept drift detection package leveraging PDP for real-time monitoring. Our study demonstrates its efficiency in detecting concept drift while providing explanations for model changes. To our knowledge, **datadriftR** is the first package combining explainable and widely used drift detection methods.

The rest of the paper is structured as follows: the next section discusses data drifts, followed by the concept drift detection methods in the package. We then compare their performance on synthetic and real-world datasets, concluding with future research directions.

2 Preliminaries

Let $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ be a predictive model with parameters θ , trained on $D = \{(x_i, y_i)\}_{i=1}^n$ to minimize loss $L(f_\theta(x_i), y_i)$. The optimal parameters $\hat{\theta}$ are found by solving $\hat{\theta} = \arg \min_{\theta} \sum_{i=1}^n L(f_\theta(x_i), y_i)$. Data drift occurs when the training distribution $P_{\text{train}}(X, Y)$ differs from the test distribution $P_{\text{test}}(X, Y)$, impacting model performance.

Covariate drift. The explanatory variable distribution changes while the response variable remains the same: $P_{\text{train}}(X) \neq P_{\text{test}}(X)$, causing prediction difficulties despite good training performance.

Label drift. The response variable distribution changes while the explanatory variable remains constant: $P_{\text{train}}(Y) \neq P_{\text{test}}(Y)$, leading to increased misclassification, especially in imbalanced datasets.

Concept drift. Both explanatory and response variable distributions change: $P_{\text{train}}(X, Y) \neq P_{\text{test}}(X, Y)$, reducing model accuracy and reliability.

The next section describes methods for detecting these drifts.

3 Methods

This section describes the existing drift detection methods such as Drift Detection Method, Early Drift Detection Method, Hoeffding Drift Detection Methods, Kolmogorov-Smirnov test-based Windowing and Page-Hinkley, as well as our proposed concept drift detection method Profile Drift Detection.

Hoeffding's Drift Detection Method on Average

HDDM-A detects abrupt concept drifts in data streams based on Hoeffding's Inequality. It monitors a simple moving average and evaluates changes using confidence levels α_D (for drift detection) and α_W (for alerts).

The data stream is divided into two windows: **minimum** ($n_{\text{min}}, c_{\text{min}}$) and **maximum** ($n_{\text{max}}, c_{\text{max}}$), which track past and current states, updating when necessary.

¹<https://www.evidentlyai.com/>

²<https://www.seldon.io/>

³<https://nannyml.readthedocs.io/en/stable/index.html>

⁴<https://frouros.readthedocs.io/en/latest/>

⁵<https://riverml.xyz/0.8.0/examples/concept-drift-detection/>

Error bounds for drift detection are computed as:

$$\epsilon_{\alpha_D} = \sqrt{\frac{1}{2n} \ln \left(\frac{1}{\alpha_D} \right)}, \quad \epsilon_{\alpha_{D1}} = \sqrt{\frac{1}{2n_{\min}} \ln \left(\frac{1}{\alpha_D} \right)}, \quad \epsilon_{\alpha_{D2}} = \sqrt{\frac{1}{2n_{\max}} \ln \left(\frac{1}{\alpha_D} \right)} \quad (1)$$

Windows update if:

$$\frac{c_{\min}}{n_{\min}} + \epsilon_{\alpha_{D1}} \geq \frac{c}{n} + \epsilon_{\alpha_D} \Rightarrow (c_{\min}, n_{\min}) \leftarrow (c, n) \quad (2)$$

$$\frac{c_{\max}}{n_{\max}} - \epsilon_{\alpha_{D2}} \leq \frac{c}{n} - \epsilon_{\alpha_D} \Rightarrow (c_{\max}, n_{\max}) \leftarrow (c, n) \quad (3)$$

Drift is detected when:

$$\left| \frac{c}{n} - \frac{c_i}{n_i} \right| \geq \sqrt{\frac{n - n_i}{n_i} \frac{1}{2n} \ln \left(\frac{1}{\alpha} \right)}, \quad i \in \{\min, \max\} \quad (4)$$

Using $\alpha = \alpha_D$ detects drift, while $\alpha = \alpha_W$ triggers an alert state.

Hoeffding's Drift Detection Method on Weighted

HDDM-W detects concept drifts in data streams using Exponentially Weighted Moving Averages (EWMA) and Hoeffding's Inequality. The EWMA statistic is computed as:

$$\hat{\mu}_{\text{EWMA}} = \lambda x_i + (1 - \lambda) \hat{\mu}_{\text{EWMA}} \quad (5)$$

where $\lambda \in [0, 1]$ controls the weight of recent observations.

HDDM-W is better suited for detecting gradual drifts, while HDDM-A excels at detecting abrupt drifts. Like DDM and EDDM, implementations in `scikit-multiflow` and `river` detect drift based on prediction errors, without assuming any specific data distribution.

Kolmogorov-Smirnov Windowing Method

KSWIN is a non-parametric drift detection method based on the Kolmogorov-Smirnov (KS) test (Raab et al., 2020). It maintains a sliding window Ψ of n observations, splitting it into:

- R : the most recent r observations,
- W : r randomly selected older observations.

The KS test computes the maximum absolute difference between empirical distributions:

$$\text{dist}_{w,r} = \sup_x |F_W(x) - F_R(x)| \quad (6)$$

Drift is detected if:

$$\text{dist}_{w,r} > \sqrt{-\frac{\ln \alpha}{r}} \quad (7)$$

KSWIN detects drift without distributional assumptions and applies to both response and explanatory variables, identifying changes even before response values are observed.

Page-Hinkley Test

The Page-Hinkley Test (PH) detects mean shifts in data streams, similar to Cumulative Sums (Sebastião and Fernandes, 2017). It incorporates a forgetting factor α to reduce the impact of older data. Initially, $n = 1$, $x_{\text{ort}} = 0$ and $S = 0$. For each new observation:

$$x_t = x_{\text{ort}} + \frac{x_t - x_{t-1}}{n} \quad (8)$$

$$S_t = \max(0, \alpha \cdot S_{t-1} + x_t - x - \delta) \quad (9)$$

If $S_t > \lambda$, a concept drift is detected.

PH can be applied to explanatory or response variables and does not require waiting for the response variable's actual value. Unlike KSWIN, PH does not include an alert state.

Drift Detection Method (DDM)

DDM estimates error rate variability using:

$$s_i = \sqrt{\frac{p_i(1-p_i)}{i}} \quad (10)$$

For large samples, the error rate follows a normal distribution with a confidence interval:

$$p_i \pm \alpha \cdot s_i \quad (11)$$

Warning Level:

$$p_i + s_i \geq p_{\min} + 2 \cdot s_{\min} \quad (12)$$

Drift Level:

$$p_i + s_i \geq p_{\min} + 3 \cdot s_{\min} \quad (13)$$

DDM is model-independent and efficient but does not detect virtual drift in explanatory variables.

Early Drift Detection Method (EDDM)

EDDM extends DDM by tracking error distances (Baena-Garcia et al., 2006). It computes:

$$p'_i + 2 \cdot s'_i \quad (14)$$

and updates $(p'_{\max}, s'_{\max})$ accordingly.

Warning Level:

$$\frac{p'_i + 2 \cdot s'_i}{p'_{\max} + 2 \cdot s'_{\max}} < 0.95 \quad (15)$$

Drift Level:

$$\frac{p'_i + 2 \cdot s'_i}{p'_{\max} + 2 \cdot s'_{\max}} < 0.90 \quad (16)$$

When drift occurs, the model is retrained and $(p'_{\max}, s'_{\max})$ are reset. EDDM is better for gradual drifts but, like DDM, cannot detect virtual drift.

Profile Drift Detection

The combination of L2 distance, first-order derivative distance (L2Der) and the Partial Dependence Index (PDI) enables a comprehensive assessment of PDP comparisons (Kobylińska et al., 2024).

- **L2 Distance** measures the overall difference and magnitude between two profiles. This method evaluates the structural similarity or differences between PDP shapes by considering squared differences between values across all points in the profiles.
- **L2Der** compares the derivatives of profiles to analyze the rate and severity of changes within profiles. Comparing derivatives effectively identifies differences in slopes and local changes, aiding in understanding both the general shapes and dynamic behaviors of profiles
- **PDI** focuses on whether profiles show an increase or decrease at the same data points, evaluating behavioral similarities by taking into account orientation changes between profiles.

Together, these metrics provide a holistic measure of PDP disparities, capturing distance, change rates and directional differences. The combined metric is referred to as PDD, illustrated in Figure~2.

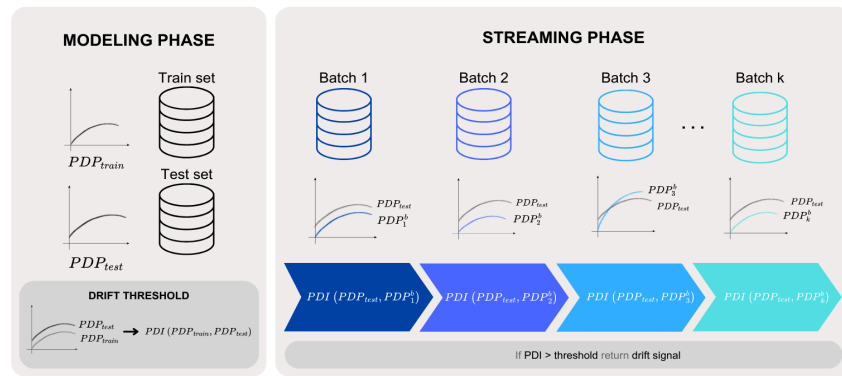


Figure 2: The workflow of the Profile Drift Detection

PDD allows parameterization of three metrics: PDI, L2 and L2Der. Previous studies (Dar and Cavus, 2023, 2024) found PDI alone insufficient for detecting concept drift. PDD enhances detection by incorporating L2 and L2Der, with thresholds set to allow acceptable deviations.

4 Using **datadriftR** package

The **datadriftR** package detects concept drift in real-time data streams using various statistical methods, including DDM, EDDM, HDDM-A, HDDM-W, KSWIN, PH and PDD. These methods are based on established research and can be applied in both dynamic and static data environments.

The package is implemented using the R6 object-oriented paradigm, enhancing flexibility and encapsulation for complex data scenarios.

Below is an example of generating a data stream and applying DDM for drift detection using **datadriftR**:

```
> library(datadriftR)

# Generate a sample data stream of 1000 elements with equal probabilities for 0 and 1
> set.seed(123) # Setting a seed for reproducibility
> data_part1 <- sample(c(0, 1), size = 500, replace = TRUE, prob = c(0.7, 0.3))

# Introduce a change in data distribution
> data_part2 <- sample(c(0, 1), size = 500, replace = TRUE, prob = c(0.3, 0.7))

# Combine the two parts
> data_stream <- c(data_part1, data_part2)

# Initialize the DDM object
> ddm <- DDM$new()

# Iterate through the data stream
> for (i in seq_along(data_stream)) {
  ddm$add_element(data_stream[i])
  if (ddm$change_detected) {
    message(paste("Drift detected!", i))
  } else if (ddm$warning_detected) {
    # message(paste("Warning detected at position:", i))
  }
}

Drift detected! 560
```

This example demonstrates initializing the DDM detector, processing a simulated data stream and detecting drift at position 560. The same approach applies to EDDM, HDDM-A, HDDM-W, KSWIN and PH by initializing their respective classes.

The ProfileDifference class calculates differences between profiles using various methods, including PDD. Below is an example:

```

> profile1 <- list(x = seq(0, 10, length.out = 100), y = sin(seq(0, 10, length.out = 100)))
> profile2 <- list(x = seq(0, 10, length.out = 100), y = cos(seq(0, 10, length.out = 100)))
> pdd <- ProfileDifference$new(method = "pdi", deriv = "gold")

> pdd$set_profiles(profile1, profile2)
> result <- pdd$calculate_difference()
> print(result$distance)
[1] 0.5154639

```

This method is adaptable to various configurations, making ProfileDifference a flexible tool for profile comparisons in dynamic data environments.

5 Experiments

In this section, traditional methods and the PDD method are compared using synthetic and real-world datasets frequently used. The characteristics of the benchmark dataset and the design of the experiment are described.

Datasets

In the experiments, we used the generic synthetic and real-world datasets SEA (Street and Kim, 2001), Hyperplane (Hulten et al., 2001), NOAA (Elwell and Polikar, 2011), Ozone (Zhang and Fan, 2008), Elec2 (Harries et al., 1999) and Friedman (Ikononovska et al., 2011). The characteristics of these datasets are provided in Table~1.

Table 1: The characteristics of the benchmark dataset

Dataset	Type	Task	#observations	#variables
SEA	Synthetic	Classification	50000	4
Hyperplane	Synthetic	Classification	20000	3
NOAA	Real	Classification	18159	9
Ozone	Real	Classification	2534	73
Elec2	Real	Classification	45312	9
Friedman	Synthetic	Regression	20000	11

Design of experiments

Experiments were conducted using Logistic Regression (LR), Decision Tree (DT) and Random Forest (RF). For the Friedman dataset, Linear Regression replaced LR.

For each incoming batch, its suitability for training was evaluated using PDI, L2Der and L2 calculated from PDPs. If all three exceeded their thresholds, the batch was added to the training set and the model was retrained. When PDD detected concept drift, PDP curves for train-test sets and graphs at the drift point were recorded, ensuring continuous model adaptation.

Statistical tests verified whether the response variable was correctly predicted. If drift was detected, similar incoming data was added to the training set, followed by retraining.

Tables~2-7 summarize the drift detection methods, batch numbers, test-train sizes, detected drifts and model performance. Large batch sizes may miss drifts, while small sizes may cause false detections, making optimal batch size selection critical.

Additional details on batch sizes (Table~8), key variables (Table~9) and model accuracy (Table~10) are provided end of the paper.

Results

In this section, the results of experiments conducted on the datasets—SEA, Hyperplane, NOAA, Ozone, Elec2 and Friedman—are summarized. The methods' performance is evaluated based on accuracy and the number of detected drifts (#drifts).

SEA

The PDI metric was excluded due to shape similarity and L2/L2Der cutoff values were set as five times the test set values in logistic regression model in this experiment. Table~2 evaluates the performance of drift detection methods across models. PDD generally achieved high accuracy, particularly with 10 batches, where LR and RF performed competitively or better than other methods. KSWIN and EDDM also showed high accuracy but detected significantly more drifts, indicating potential over-sensitivity. HDDM-A and HDDM-W detected fewer drifts, especially with smaller batches, while maintaining accuracy comparable to PDD. PH demonstrated stable drift detection but had lower accuracy than PDD. DDM either failed to detect drifts or detected very few, suggesting lower sensitivity.

Table 2: The results of experiments on the SEA dataset

Method	Model	#batch					
		10		20		30	
		accuracy	#drifts	accuracy	#drifts	accuracy	#drifts
HDDM-A	LR	0.8306	1	0.8462	2	0.8447	2
	DT	0.8318	2	0.8400	2	0.8358	2
	RF	0.8264	1	0.8432	2	0.8384	2
HDDM-W	LR	0.8279	3	0.8443	2	0.8407	2
	DT	0.8248	3	0.8389	2	0.8354	2
	RF	0.8257	3	0.8412	4	0.8362	3
KSWIN	LR	0.8292	10	0.8455	20	0.8428	27
	DT	0.8294	10	0.8440	20	0.8376	29
	RF	0.8268	10	0.8421	20	0.8387	29
PH	LR	0.8337	1	0.8447	3	0.8437	3
	DT	0.8307	1	0.8424	3	0.8368	3
	RF	0.8298	1	0.8417	3	0.8393	3
DDM	LR	0.8284	0	0.8460	3	0.8426	5
	DT	0.8267	0	0.8432	1	0.8356	3
	RF	0.8267	0	0.8430	3	0.8379	3
EDDM	LR	0.8292	10	0.8455	20	0.8381	12
	DT	0.8291	9	0.8435	18	0.8340	8
	RF	0.8268	10	0.8423	19	0.8390	13
PDD	LR	0.8309	5	0.8464	18	0.8370	1
	DT	0.8230	1	0.8405	3	0.8349	8
	RF	0.8283	5	0.8427	8	0.8377	18

PDD detected drifts in a more controlled manner, reducing false positives and ensuring stable performance, especially in medium and large batch sizes. In contrast, KSWIN and EDDM detected more drifts, increasing false positives, while HDDM-A and HDDM-W detected fewer, potentially missing actual changes. PH and DDM showed moderate detection, varying by drift characteristics.

By model type, PDD maintained high accuracy in LR while detecting fewer drifts for stability. In DT and RF, it provided consistent results with fewer detections than other methods. RF showed high accuracy but KSWIN and EDDM detected more drifts, increasing operational noise.

PDD's advantages include reducing false positives, maintaining accuracy across models and batch sizes and ensuring stability, making it reliable for evolving data streams.

LR detected five drifts in the 10th batch. As shown in Figure~3, PDPs between train and test sets remained similar, indicating consistent model performance. Even after drift detection, PDPs retained their shape. PDI was 0 at drift points, confirming stable functional relationships despite distribution shifts.

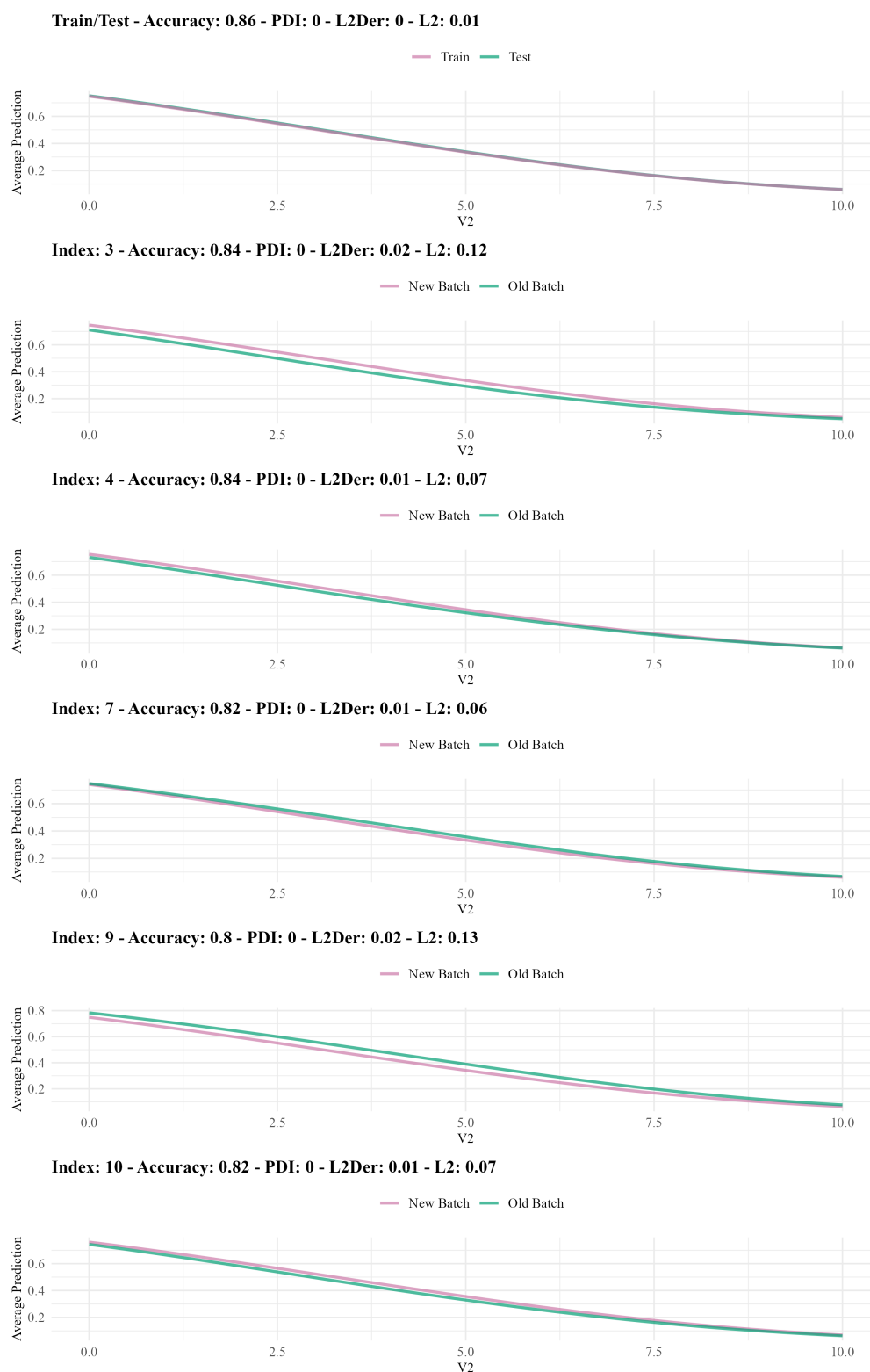


Figure 3: PDPs of the Logistic Regression model trained on SEA dataset

For the DT model, a drift was detected at the 4th index. As shown in Figure~4, variable V1 significantly contributed to this drift. Specifically, within the range of 0 to 5, V1's influence on the response variable declined, indicating a weakened predictive power. This suggests a shift in the relationship between V1 and the target variable, likely due to changes in data distribution. Such shifts emphasize the need to monitor variable contributions over time, as subtle alterations can signal drifts that may impact model performance.

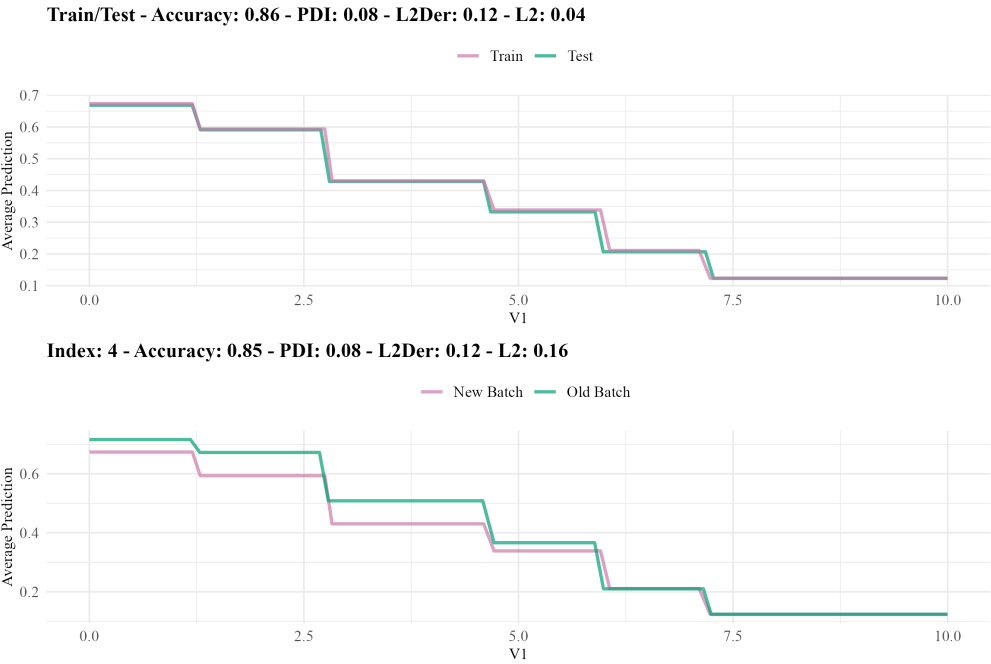


Figure 4: PDPs of the Decision Tree model trained on SEA dataset

Variable V2 was identified as the most important feature in the RF model. Figure~5 shows that V2 played a key role at the 3rd and 10th indices. Within the range of 0 to 7.5, noticeable variations in V2’s behavior suggest a shift in its influence on the model’s output, likely due to data distribution changes in these batches. These shifts highlight how variations in key variables can impact model performance and generalization. Monitoring important variables is crucial, as small changes can trigger drift detection and necessitate model adaptation, especially in real-time data streams.

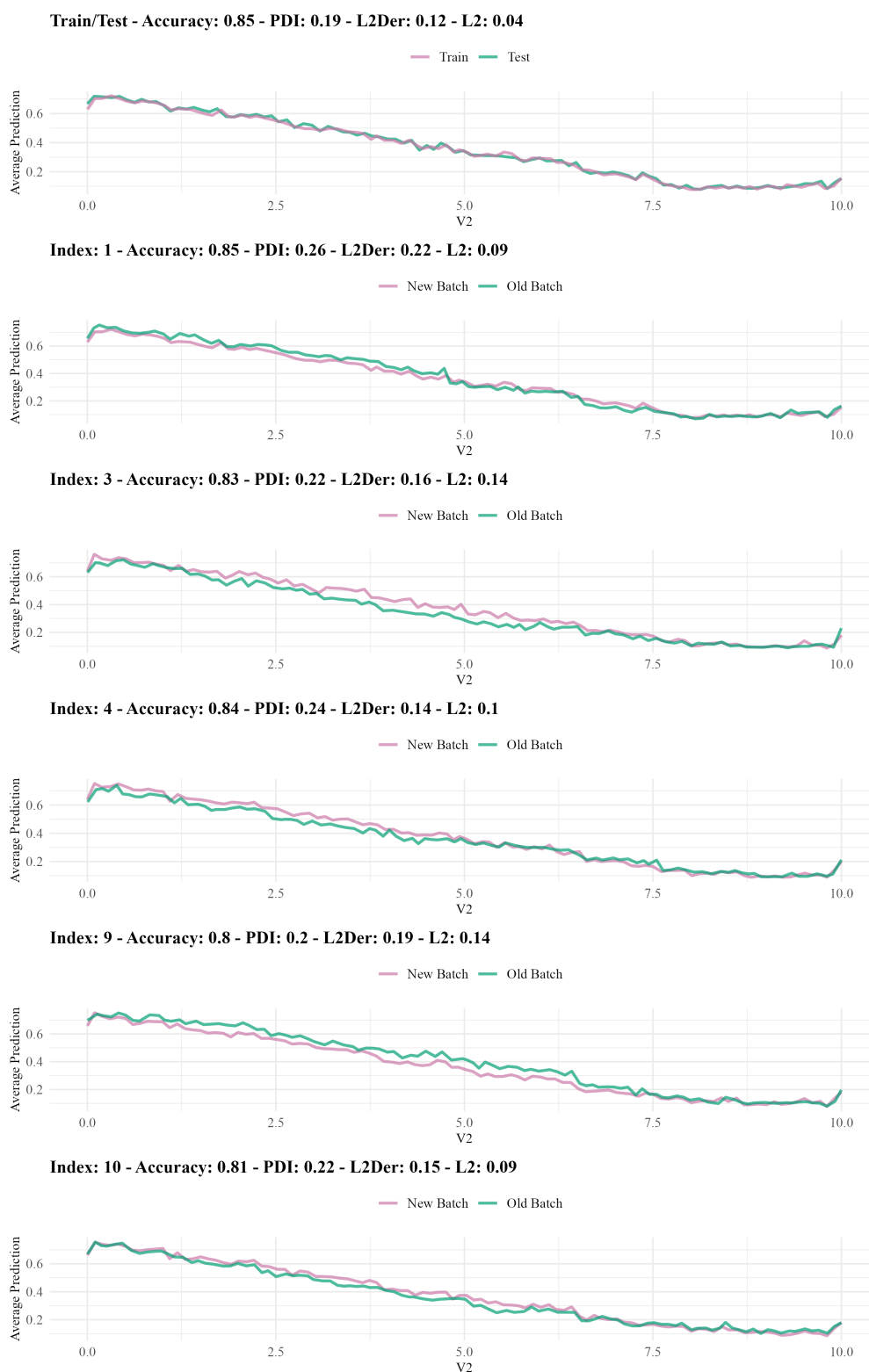


Figure 5: PDPs of the Random Forest model trained on SEA dataset

Hyperplane

Table~3 shows that for the Hyperplane dataset, the PDD method achieved competitive accuracy at 10 and 20 batches in DT and RF models. The LR model reached 0.8648 accuracy with 10 batches, comparable to KSWIN and EDDM. However, as batches increased to 30, PDD's accuracy dropped significantly to 0.5219 for LR, indicating reduced performance beyond 20 batches.

Table 3: The results of experiments on the Hyperplane dataset

Method	Model	#batch					
		10		20		30	
		accuracy	#drifts	accuracy	#drifts	accuracy	#drifts
HDDM-A	LR	0.8662	1	0.8204	2	0.7568	3
	DT	0.8398	0	0.7670	3	0.6921	3
	RF	0.7834	0	0.6809	1	0.6267	1
HDDM-W	LR	0.8377	1	0.8456	3	0.8152	4
	DT	0.8398	0	0.7752	2	0.7316	4
	RF	0.7794	2	0.7615	3	0.7536	6
KSWIN	LR	0.8639	9	0.8458	13	0.8285	18
	DT	0.8520	9	0.8429	17	0.8162	19
	RF	0.8219	10	0.8144	17	0.7948	22
PH	LR	0.8648	0	0.8187	1	0.7455	2
	DT	0.8398	0	0.7014	1	0.6929	1
	RF	0.7834	0	0.7033	1	0.6267	1
DDM	LR	0.8648	0	0.8071	1	0.7455	2
	DT	0.8398	0	0.7014	1	0.7283	3
	RF	0.7834	0	0.7290	3	0.6267	1
EDDM	LR	0.8648	0	0.8485	13	0.8348	5
	DT	0.8496	10	0.8562	15	0.8379	11
	RF	0.7834	0	0.8072	16	0.7957	24
PDD	LR	0.8648	0	0.6469	1	0.5219	0
	DT	0.8435	1	0.8326	15	0.4924	1
	RF	0.7831	1	0.7322	4	0.5099	0

An analysis of train-test accuracies suggests overfitting, particularly in RF. For LR, training accuracy was 0.7320, while test accuracy dropped from 0.6165 (10 batches) to 0.5113 (30 batches). Similar trends in DT and RF indicate overfitting, likely impacting PDD's drift detection.

PDD identified fewer drifts under overfitting, whereas KSWIN and EDDM detected up to 22. This suggests PDD reduces false positives but may miss actual drifts in overfitting scenarios.

KSWIN and EDDM detected more drifts, increasing sensitivity but also false positives. HDDM-A and HDDM-W detected fewer drifts, balancing sensitivity and stability. PH showed moderate detection, while DDM performed inconsistently, especially in RF.

Overall, PDD balanced accuracy and drift detection in the Hyperplane dataset, especially at 20 batches. However, its performance declined outside this range due to overfitting. While PDD minimizes false positives, its sensitivity to actual drifts may be limited when model generalization is affected.

NOAA

The results for the NOAA dataset in Table~4 show that PDD performed effectively mainly at 20 batches. For LR, accuracy increased from 0.7649 (10 batches) to 0.7708 (30 batches) but drift detections were minimal, with 0 drifts at 10 and 30 batches and only 2 at 20 batches.

Table 4: The results of experiments on the NOAA dataset

Method	Model	#batch					
		10		20		30	
		accuracy	#drifts	accuracy	#drifts	accuracy	#drifts
HDDM-A	LR	0.7736	5	0.7696	7	0.7777	3
	DT	0.7293	8	0.7435	9	0.7440	8
	RF	0.7790	0	0.7783	4	0.7780	2
HDDM-W	LR	0.7736	7	0.7712	12	0.7799	12
	DT	0.7385	10	0.7345	16	0.7385	23
	RF	0.7883	4	0.7835	10	0.7826	9
KSWIN	LR	0.7755	10	0.7755	20	0.7788	27
	DT	0.7385	10	0.7334	20	0.7437	29
	RF	0.7916	10	0.7916	19	0.7945	29
PH	LR	0.7707	3	0.7680	3	0.7792	3
	DT	0.7402	2	0.7387	2	0.7276	2
	RF	0.7869	3	0.7798	3	0.7772	3
DDM	LR	0.7716	4	0.7746	10	0.7715	10
	DT	0.7422	3	0.7334	20	0.7398	21
	RF	0.7818	1	0.7914	20	0.7831	8
EDDM	LR	0.7755	10	0.7755	20	0.7780	30
	DT	0.7385	10	0.7334	20	0.7406	30
	RF	0.7916	10	0.7914	20	0.7940	30
PDD	LR	0.7649	0	0.7694	2	0.7708	0
	DT	0.7264	2	0.7403	0	0.7352	3
	RF	0.7881	5	0.7676	0	0.7913	22

Overfitting is a primary issue in the RF model, with train accuracy reaching 0.9998 while test accuracy ranges from 0.7723 to 0.8338. In contrast, LR and DT show smaller train-test gaps, indicating minimal overfitting.

PDD detected fewer drifts in overfitting scenarios, especially in RF, identifying 22 drifts at 30 batches compared to 29 and 30 detected by KSWIN and EDDM. While reducing false positives, PDD risks missing actual drifts in overfitting conditions.

Other methods varied: KSWIN and EDDM detected more drifts but increased false positives, HDDM-A and HDDM-W balanced sensitivity and stability, PH detected fewer drifts and DDM showed moderate sensitivity.

Overall, PDD showed limited effectiveness in the NOAA dataset. KSWIN and EDDM provided more consistent drift detection but at the cost of higher false positives.

Ozone

The results for the Ozone dataset, shown in Table~5, highlight varied performances across drift detection methods, models and batch sizes. The PDD method achieved moderate accuracy, with 0.9353 at 10 batches, 0.8984 at 20 batches and 0.8724 at 30 batches in the LR model. While PDD maintained a consistent number of detected drifts for LR, its performance slightly declined as batch size increased.

Table 5: The results of experiments on the Ozone dataset

Method	Model	#batch					
		10		20		30	
		accuracy	#drifts	accuracy	#drifts	accuracy	#drifts
HDDM-A	LR	0.9294	0	0.8942	2	0.8116	0
	DT	0.9487	0	0.9101	1	0.9036	2
	RF	0.9508	0	0.9427	1	0.9374	0
HDDM-W	LR	0.9294	0	0.8907	1	0.8526	1
	DT	0.9487	0	0.9101	1	0.9122	2
	RF	0.9508	0	0.9427	1	0.9401	1
KSWIN	LR	0.9396	3	0.9164	2	0.9144	5
	DT	0.9444	3	0.9242	4	0.9099	5
	RF	0.9529	1	0.9383	4	0.9437	3
PH	LR	0.9294	0	0.8502	0	0.8116	0
	DT	0.9487	0	0.8911	0	0.8725	0
	RF	0.9508	0	0.9422	0	0.9374	0
DDM	LR	0.9406	4	0.9155	6	0.8706	2
	DT	0.9358	7	0.9301	6	0.9329	4
	RF	0.9513	2	0.9427	2	0.9437	3
EDDM	LR	0.9396	2	0.9183	4	0.8842	6
	DT	0.9412	2	0.9164	4	0.9185	5
	RF	0.9524	2	0.9441	3	0.9437	3
PDD	LR	0.9353	1	0.8984	2	0.8724	2
	DT	0.9487	0	0.8915	1	0.9140	10
	RF	0.9508	5	0.9422	0	0.9455	12

In the DT model, PDD detected one drift at 20 batches, rising to ten at 30. For RF, it identified five drifts at 10 batches, none at 20 and twelve at 30, suggesting sensitivity depends on model complexity.

The target distribution (2374 to 160) impacts drift detection. At 30 batches, PDD detected more drifts than other methods, achieving the highest accuracy (0.9455). Unlike response-based methods, PDD provides deeper insights via PDP comparisons.

Other methods varied in performance: DDM, KSWIN and EDDM detected more drifts but increased false positives, while HDDM-A and HDDM-W balanced sensitivity and stability with fewer detections.

Key variables shifted across models and batch sizes, highlighting the need for adaptive selection. V26 was crucial in LR at 10 batches, shifting to V52 at 20/30; DT's key variable changed from V31 to V36 and RF's from V56 to V61.

In summary, PDD maintained accuracy while detecting fewer drifts, particularly in overfitting-prone RF models. It reduced false positives but may miss key drifts, whereas KSWIN and EDDM offered more consistent detection at the cost of increased false alarms. Adaptive variable selection and a combination of methods may enhance drift detection.

Elec2

The results for the Elec2 dataset highlight distinct performance patterns across drift detection methods, models and batch sizes. The PDD method demonstrated a balanced approach in accuracy and drift detection.

For the Logistic Regression (LR) model, PDD achieved accuracies of 0.6740, 0.7349 and 0.7402 for 10, 20 and 30 batches, respectively. In the DT model, accuracies remained consistent at 0.7253, 0.7392 and 0.7449, while in the RF model, accuracies were 0.7093, 0.7311 and 0.7393 across the same batch sizes.

Table 6: The results of experiments on the Elec2 dataset

Method	Model	#batch					
		10		20		30	
		accuracy	#drifts	accuracy	#drifts	accuracy	#drifts
HDDM-A	LR	0.6983	10	0.6836	20	0.7099	30
	DT	0.7255	10	0.7409	20	0.7450	30
	RF	0.7029	10	0.7243	20	0.7374	30
HDDM-W	LR	0.6983	10	0.6836	20	0.7099	30
	DT	0.7255	10	0.7409	20	0.7450	30
	RF	0.7029	10	0.7243	20	0.7374	30
KSWIN	LR	0.6983	10	0.6836	20	0.7099	30
	DT	0.7255	10	0.7409	20	0.7450	30
	RF	0.7029	10	0.7243	20	0.7374	30
PH	LR	0.6983	10	0.7342	13	0.7390	16
	DT	0.7215	9	0.7417	15	0.7502	16
	RF	0.6992	8	0.7103	11	0.7368	16
DDM	LR	0.6983	10	0.6836	20	0.7099	30
	DT	0.7255	10	0.7409	20	0.7450	30
	RF	0.7029	10	0.7250	16	0.7253	17
EDDM	LR	0.6983	10	0.6836	20	0.7099	30
	DT	0.7255	10	0.7409	20	0.7450	30
	RF	0.7029	10	0.7243	20	0.7374	30
PDD	LR	0.6740	8	0.7349	4	0.7402	12
	DT	0.7253	1	0.7392	1	0.7429	5
	RF	0.7093	3	0.7311	0	0.7449	3

When comparing PDD to other drift detection methods, it exhibits a moderate number of drift detections. In the LR model, PDD detected 8, 4 and 12 drifts across 10, 20 and 30 batches, respectively. In DT, drift detections were minimal, with 1 drift at 10 and 20 batches, increasing to 5 at 30 batches. The RF model showed 3, 0 and 3 drifts for the same batch sizes. This controlled detection suggests that PDD effectively identifies significant changes while minimizing false positives.

Other methods, such as HDDM-A, HDDM-W, KSWIN and EDDM, detected drifts equal to the batch size across all models. PH exhibited an increasing trend, detecting up to 16 drifts in all models at 30 batches. DDM showed a rising number of drifts, particularly in RF, indicating high sensitivity.

Accuracy-wise, most methods maintained strong performance across models and batch sizes. The LR model consistently achieved accuracies above 0.68, while DT and RF models surpassed 0.70-0.74. High base test and train accuracies indicate overall robust model performance.

A key finding in the Elec2 dataset is that `nswprice` was the most influential variable across all models and batch sizes, emphasizing its critical role in predictive performance. With 30 batches, RF achieved its highest accuracy when updated three times using PDD. Other methods detected more drifts but did not match this accuracy.

In summary, PDD demonstrated balanced performance on Elec2, maintaining accuracy while controlling drift detections. Its conservative approach reduces false positives, making it advantageous in scenarios requiring stable drift detection. The identification of `nswprice` as the key variable underscores the need for focused variable analysis in drift detection and model evaluation. Overall, PDD provides a reliable balance between accuracy and drift sensitivity.

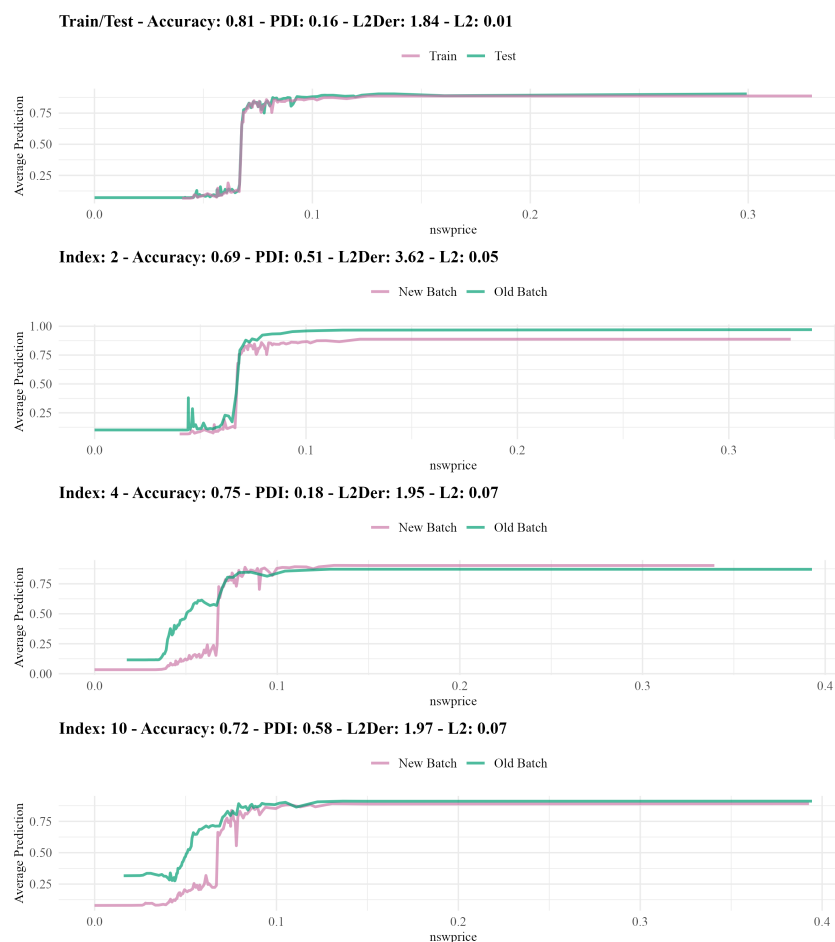


Figure 6: PDPs of the Random Forest model trained on Elec2 dataset

For the RF model applied to the Elec2 dataset, drift was detected in the 10th batch. As shown in Figure~6, the `nswprice` variable, the most important feature, exhibits noticeable changes between 0 and 0.1. In this range, its contribution to the model's predictions shifts, triggering drift detection by PDD. These variations indicate how `nswprice` influences the model differently in the 10th batch, potentially affecting its performance.

Friedman

The results for the Friedman dataset in Table~7, a regression task, show distinct performance patterns across drift detection methods, models and batch sizes. The PDD method demonstrated a balanced approach in both metric values and drift detection. The PDI metric was excluded due to shape similarity in this experiment.

Table 7: The results of experiments on the Friedman dataset

Method	Model	#batch					
		10		20		30	
		accuracy	#drifts	accuracy	#drifts	accuracy	#drifts
KSWIN	LR	2.8029	10	2.8007	19	2.7906	30
	DT	3.2977	10	3.2765	19	3.2921	23
	RF	1.9021	10	1.9280	19	1.9320	26
PH	LR	2.8023	5	2.7974	10	2.7812	9
	DT	3.3254	9	3.2813	16	3.2752	16
	RF	1.9613	2	2.0566	4	2.0115	7
PDD	LR	2.8041	3	2.8073	2	2.7998	1
	DT	3.3014	3	3.2936	2	3.2670	2
	RF	1.9887	7	1.8811	4	1.9355	11

KSWIN detected the highest number of drifts across all models and batch sizes. In LR, it identified 10, 19 and 30 drifts, with RMSE slightly decreasing from 2.8029 to 2.7906. Similarly, DT and RF models showed 10, 19 and 23 drifts and 10, 20 and 24 drifts, respectively, with minor RMSE improvements. While highly sensitive, KSWIN's frequent detections increase false positive risks.

PH showed moderate sensitivity, detecting fewer drifts than KSWIN but more than PDD. In LR, PH detected 5, 10 and 9 drifts, improving RMSE from 2.8023 to 2.7812. DT detected 9, 16 and 16 drifts, while RF detected 2, 4 and 7, maintaining stable metrics. PH balances sensitivity and stability, making it suitable for controlled drift detection.

Across all methods, `variable_4` consistently emerged as the key feature, emphasizing its importance in model performance and drift detection.

In summary, PDD controlled drift detections while maintaining reasonable metrics in the Friedman dataset. It reduced false positives but was less sensitive than KSWIN, which detected more drifts but increased false alarms. PH offered a balanced approach, while the stability of `variable_4` highlights the need for adaptive variable selection in drift detection.

6 Conclusions

In this paper, we conducted six experiments on real and synthetic datasets to compare PDD with common drift detection methods (HDDM-A, HDDM-W, KSWIN, PH, DDM, EDDM). We analyzed model accuracy, detected drifts and variable contributions during drift events.

PDD evaluates three metrics—PDI, L2 and L2Der—between PDP curves. Thresholds are set using train-test PDPs and new batch metrics are compared against them. Experiments also tested alternative conditions, such as exceeding thresholds in at least two metrics. Statistical tests and PDD were compared across batch sizes in terms of drift detection and model performance.

Beyond detecting drift, PDD provides insights into model-variable relationships. Unlike traditional methods that rely on response prediction, PDD detects changes even at high accuracy levels, allowing updates when data relationships shift without affecting standard metrics.

PDD balanced drift detection and accuracy, avoiding excessive false positives. It performed stably on SEA and Elec2, while in Hyperplane, NOAA and Ozone, overfitting impacted its drift detection. In Friedman, PDD effectively minimized false positives while maintaining accuracy.

Comparatively, KSWIN and EDDM were highly sensitive, detecting more drifts but increasing false positives. PH and DDM offered moderate detection, balancing stability and sensitivity.

Key variables consistently influenced drift detection, such as `V2` in SEA, `nswprice` in Elec2 and `variable_4` in Friedman, underscoring the need for monitoring influential attributes in dynamic data environments.

7 Discussion

Beyond detecting drift, PDD provides insights into predictive churn, helping explain model changes after retraining (Watson-Daniels et al., 2024). Unlike statistical methods, PDD clarifies model-data

relationships, aiding users in understanding drift causes. It can also contribute to data privacy challenges, such as removing learned relationships from models (Bourtole et al., 2020).

PDD has limitations, including computational cost and infeasibility for multi-class classification. Derivative calculations are difficult for categorical variables and relying on the most important variable may overlook drifts from other features, requiring broader analysis and increasing computational demands. Sparse vectors can also exaggerate PDI metric variations.

PDD is suited for binary classification and regression but adapting it for multi-class tasks is costly.

In conclusion, PDD balances drift detection with model interpretability. It detects fewer but more consistent drifts while maintaining accuracy, offering deeper insights into how explanatory variables influence concept drift.

8 Further research

To reduce PDD's computational cost, the *compress then explain* technique (Baniecki et al., 2024) can be integrated, compressing large datasets before applying explainable methods. When model training is costly, PDD can be used as is. If PDP generation is more expensive than model training, PDD can track changes in response-explanatory variable relationships. Additionally, PDD aids in predictive churn and data privacy by explaining changes after ML model retraining.

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Supplemental materials

The materials for reproducing the experiments performed and the dataset are accessible in the following repository: https://github.com/ugurdar/datadrift_RJournal

Table 8: The number of observations in the batches

Dataset	#batch					
	10		20		30	
	train	test	train	test	train	test
SEA	13332	3334	8000	2000	5713	1429
Hyperplane	5332	1334	3200	800	2285	572
NOAA	4842	1211	2904	727	2075	519
Ozone	675	169	404	102	289	73
Elec2	12083	3021	7249	1813	5178	1295
Friedman	5332	1334	3200	800	2285	572

Table 9: The most important variables in the batches

Dataset	Model	#batch		
		10	20	30
SEA	LR	V2	V2	V2
	DT	V1	V1	V1
	RF	V2	V2	V2
Hyperplane	LR	feature_1	feature_1	feature_1
	DT	feature_1	feature_1	feature_2
	RF	feature_1	feature_1	feature_1
NOAA	LR	attribute1	attribute4	attribute1
	DT	attribute2	attribute2	attribute2
	RF	attribute2	attribute4	attribute2
Ozone	LR	V26	V52	V52
	DT	V31	V36	V36
	RF	V56	V61	V61
Elec2	LR	nswprice	nswprice	nswprice
	DT	nswprice	nswprice	nswprice
	RF	nswprice	nswprice	nswprice
Friedman	LR	feature_4	feature_4	feature_4
	DT	feature_4	feature_4	feature_4
	RF	feature_4	feature_4	feature_4

Table 10: The accuracies of the models in the batches

Dataset	Model	#batch					
		10		20		30	
		test	train	test	train	test	train
SEA	LR	0.8579	0.8966	0.8946	0.9032	0.9007	0.9057
	DT	0.8576	0.8801	0.8696	0.8850	0.8797	0.8850
	RF	0.8537	0.9991	0.8896	0.9989	0.9007	0.9991
Hyperplane	LR	0.6165	0.7320	0.6717	0.7122	0.5113	0.7580
	DT	0.7079	0.7406	0.7266	0.7094	0.4276	0.7575
	RF	0.6524	0.9992	0.6692	0.9984	0.4799	0.9978
NOAA	LR	0.7748	0.7999	0.8118	0.8041	0.8173	0.8014
	DT	0.7500	0.7881	0.7734	0.7996	0.7808	0.7990
	RF	0.7723	0.9996	0.8338	0.9997	0.8154	0.9995
Ozone	LR	0.9647	0.9526	0.8155	1.0000	0.9054	1.0000
	DT	0.9882	0.9511	0.8544	0.9554	0.9730	0.9308
	RF	0.9941	1.0000	0.9806	1.0000	1.0000	1.0000
Elec2	LR	0.8154	0.8200	0.8043	0.8081	0.8156	0.8188
	DT	0.7955	0.8264	0.8649	0.8238	0.8025	0.8316
	RF	0.8107	0.8424	0.8142	0.8367	0.8040	0.8418
Friedman	LR	2.6432	2.5926	2.5591	2.5579	2.5958	2.5463
	DT	3.0254	2.9843	3.1574	2.9204	3.0400	2.9744
	RF	1.6395	0.7275	1.7751	0.7659	1.7454	0.8102

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